

Wonseok Oh

Ph.D Student

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Homepage

<https://marinelay.github.io>

Research Areas

My research focuses on **Programming Language** and **Software Engineering**, specifically developing advanced program analysis techniques to automatically detect bugs and guarantee software reliability in complex systems.

Education

Mar 2024 - Present

Ph.D. in Computer Science - Korea University

Mar 2022 - Feb 2024

M.S. in Computer Science - Korea University

Mar 2016 - Aug 2021

B.S. in Computer Science - Korea University (GPA 4.41 / 4.5)

Publications

1. Miryeong Kang, Wonseok Oh, Gabin An, Hakjoo Oh. *Pig: Leveraging Large Language Models for Python Library Migrations*
FSE 2026: ACM International Conference on the Foundations of Software Engineering
🏆 ACM SIGSOFT Distinguished Paper Award; 🏆 ACM SIGSOFT Distinguished Artifact Award
2. Wonseok Oh, Hyobin Park, Miryeong Kang, Seungbin Choi, Yunja Choi, Hakjoo Oh. *PySTAAR: An End-to-End, Extensible Framework for Automated Python Type Error Repair*
ICSE 2026 Demo: IEEE/ACM International Conference on Software Engineering (Demonstrations Track)
3. Wonseok Oh, Hakjoo Oh. *TypeCare: Boosting Python Type Inference Models via Context-Aware Re-Ranking and Augmentation*
ICSE 2026: IEEE/ACM International Conference on Software Engineering
🏆 ACM SIGSOFT Distinguished Artifact Award
4. Wonseok Oh, Hakjoo Oh. *Towards Effective Static Type-Error Detection for Python*
ASE 2024: 39th IEEE/ACM International Conference on Automated Software Engineering
5. Wonseok Oh, Hakjoo Oh. *PyTER: Effective Program Repair for Python Type Errors*
ESEC/FSE 2022: ACM Joint European Software Engineering Conference and Symposium on the Foundations of Software Engineering

Professional Activities

■ Artifact Evaluation Committee (AEC) Members

- SAS 2025: 32nd Static Analysis Symposium

■ Patent

- Device and method for program repair for type errors in dynamically typed language

Talks

- **Apr 2026.** ICSE 2026, *TypeCare: Boosting Python Type Inference Models via Context-Aware Re-Ranking and Augmentation*
- **Oct 2024.** ASE 2024, *Towards Effective Static Type-Error Detection for Python*
- **Feb 2023.** KCSE 2023 Invite Talk, *PyTER: Effective Program Repair for Python Type Errors*
- **Nov 2022.** FSE 2022, *PyTER: Effective Program Repair for Python Type Errors*